

RIYA SAHNI

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New York, NY

RESEARCH FOCUS

I work on Human-AI Interaction, Human-Agent Interaction, Agentic Systems, and Responsible AI. I design tools and frameworks that make AI behavior more transparent, steerable, and aligned with human intent in creative and enterprise settings. My work spans qualitative studies of AI adoption and the design of agentic systems.

EDUCATION

Columbia University

Ph.D. Candidate in Computer Science
Advisor: Dr. Lydia B. Chilton

New York, NY

The University of Chicago

B.A. in Economics
B.S. in Computer Science with a Specialization in Human-Computer Interaction,
Advisors: Dr. Marshini Chetty, Dr. Sarah Sebo

Chicago, IL

GRANTS

National Science Foundation Graduate Research Fellowship

2024

Jeff Metcalf Award

2021

WORK EXPERIENCE

Microsoft

New York, NY

Customer Success Account Manager in Banking Financial Services

August 2023 – September 2024

- Managed unified enterprise contracts for two major U.S. credit unions; led renewal strategy, service-credit planning, and success delivery with engineering and sales.
- Executed a six-month roadmap that secured a full-term renewal and aligned Microsoft services (EDE, STA, Mission Critical) to customer transformation goals.
- Directed cross-functional coordination on escalations and delivery engagements to maintain account health and demonstrate value realization.
- Elected to the Aspire Executive Leadership Team (ELET); organized the Aspire Reverse Mentorship Program and served liaison between the Aspire community and senior leadership.

Microsoft AI Ethics and Effects in Engineering and Research (AETHER)

New York, NY

Research Volunteer; Advisor: Dr. Mihaela Vorvoreanu

August 2023 – September 2024

- Constructed a codebook and qualitatively coded data from over 50 participants in a mixed-methods user-study to measure dependency between users and Copilot.
- Study results are published in Microsoft internal library and were presented at Ignite 2024.

INTERNSHIP EXPERIENCE

Microsoft

Chicago, IL

- *Data and AI Cloud Solutions Architect (CSA) Intern*
June 2022 – September 2022
 - Developed content for and delivered ~80 minutes of live demos and presentations to two external clients of ~20-25 developers and data analysts on Git and GitHub for non-developers. Clients included Allstate and Cerner. Received consistent feedback on high quality of trainings at the level of seasoned CSAs and was recommended for full-time offer post internship.
 - Developed content for and delivered live demos and presentations to internal clients on Synapse and Azure DevOps integration, linking GitHub repository with Azure to utilize Azure services.
 - Delivered Azure synapse-in-a-day with live demo on Azure pipelines to external clients.
 - Achieved Microsoft Azure Fundamentals AZ-900 certification.

- *Data and AI Customer Engineer Intern* June 2021 – September 2021
 - Developed content and delivered live demonstrations and presentations in two Microsoft Power Activates to a group of ~15 GoPro clients about creating visualizations in a shared Power BI dataset and an introduction to Quick Measures in Power BI.
 - Led live demonstrations and discussions in two separate Microsoft Chalk Talks to a group of 25-30 Humana clients about the basics of DAX formulas and basics of Power Query in Power BI.
 - Achieved Microsoft Data Associate Analyst certification.
 - Placed in the top 20% of 130 interns in the Microsoft Intern Logic Puzzle Tournament.

Tiburon Resources

Software Developer Intern

Houston, TX

January 2021 – June 2021

- Developed and debugged API calls in Python to extract energy-related information from public platforms for company use.
- Conducted and presented data analysis for global energy demand scenarios.
- Generated insights on how consumer behavior, policy frameworks, and Paris Agreement drive energy transition.

Chevron Information Technology Company

Data Engineering Intern, Data Analytics and Modern Data Teams

Houston, TX

June 2020 – September 2020

- Implemented a cloud data warehouse with Azure SQL Server and Microsoft SQL Server Management Studio, by collaborating with multiple teams using Agile/Kanban methods.
- Developed multiple Power BI reports for data analytics.
- Placed 4th out of 11 teams in Intern Data Science Team Challenge:
 - Built machine learning algorithms in Python to test regression models like XGBoost, RandomForest, Lasso, SVR, etc. on various data sets.
 - Competition objective was to maximize hot dog sales and minimize food waste/production costs by outlining an efficient selling plan for hot dog vendors across the United States.

RESEARCH PUBLICATIONS

- **Under Submission** – *AgentDynEx: Nudging the Mechanics and Dynamics of Multi-Agent Simulations*. **Riya Sahni***, Jenny Ma*, Karthik Sreedhar, Lydia B. Chilton. <https://arxiv.org/pdf/2504.09662>.
- **HICSS 2026** – *Beyond Training: How Workers Discover Value in Enterprise AI*. **Riya Sahni**, Lydia B. Chilton. https://riyasahni.github.io/papers/copilot_paper.pdf.
- **CSCW 2025** – *An Experimental Study of Netflix Use and the Effects of Autoplay on Watching Behaviors*. Brennan Schaffner, Yarezi Ulloa, **Riya Sahni**, Jiatong Li, Ava Kim Cohen, Natasha Messier, Lan Gao, Marshini Chetty. <https://dl.acm.org/doi/abs/10.1145/3710928>.
- **CHI 2025** – *DynEx: Dynamic Code Synthesis with Structured Design Exploration for Accelerated Exploratory Programming*. Jenny Ma, Karthik Sreedhar, Vivian Liu, Sitong Wang, Pedro Alejandro Perez, **Riya Sahni**, Lydia B. Chilton. (Best paper honorable mention). <https://dl.acm.org/doi/full/10.1145/3706598.3714115>.

RESEARCH EXPERIENCE

The University of Chicago, Department of Computer Science

Chicago, IL

- *Research Assistant; Advisor: Dr. Marshini Chetty, AIR Lab* February 2023 – January 2024
 - Developed Qualtrics survey questions for an extensive, multi-part user study on Netflix.
 - Recruited participants and conducted preliminary studies to iterate survey design.
 - Presented EdTech tools poster regarding online child data privacy and security study information at the National Robotics Week Block Party hosted by Chicago's Museum of Science and Industry.
- *Research Volunteer; Advisor: Dr. Sarah Sebo, HRI Lab* March 2023 – June 2023
 - Collected qualitative data by recruiting participants, developing, and conducting virtual interviews with teachers across budget-limited U.S. elementary public schools.
 - Transcribed recordings and initiated qualitative coding.

TEACHING

Columbia University

New York, NY

Computer Science Teaching Assistant for COMS 4170 UI Design

January 2025 – June 2025

- Mentored 25 students; hosted weekly office hour sessions; graded homework.

The University of Chicago

Chicago, IL

Computer Science Teaching Assistant for MPCS 53001 Databases

September 2022 – December 2022

- TA for 60 students in masters-level DB course.
- Hosted weekly office hour sessions, graded homework and exams.

SERVICE

ReadAhead, Mentorship Program

New York, NY

Reading mentor to an assigned elementary school student

October 2024 – Now

- Conduct bi-weekly reading sessions with an elementary school student, focusing on reading comprehension and vocabulary.

Columbia University, Graduate Org for Computer Science (GrOCS)

New York, NY

Founding Member and Head of Social Media and Communications

December 2024 – Now

- Manage primary social channel; event promotion.

Microsoft

New York, NY

Lead Event Programmer for Microsoft Tristate Disability ERG NY Chapter

October 2023 – September 2024

- Founded and hosted a bi-weekly lunch-and-learn series on accessibility and inclusive technology.
- Co-planned a 250+ person panel discussion with the Eagles Autism Foundation.
- Co-planned a 200+ person virtual toy drive with Holton's Heroes.

SKILLS

Programming: Python, Rust, JavaScript/Node.js, C, CSS/HTML

AI & Systems: OpenAI API, prompt engineering, M365 Copilot, Azure AI (fundamentals)

Data & Tools: Power BI, MySQL, MongoDB, pandas, Flask, Git/GitHub, Figma

Languages: English (native), conversational Spanish and Hindi